Lab 1: Set up a Cloud Account – 1BM24MC021

- **Problem Statement:** Your company plans to move infrastructure to the cloud. As a cloud engineer, set up a new cloud account.
- **Tasks:**
  1. Register for AWS/Azure/GCP Free Tier.
  2. Explore dashboard services.
  3. Note down free tier limitations.

Lab 2: Deploy a Static Website - 1BM24MC005

- **Problem Statement:** A startup wants to publish a company profile webpage.
- **Tasks:**
  1. Create an S3/Blob bucket.
  2. Upload HTML/CSS files.
  3. Enable static website hosting.

Lab 3: Cloud Service Models Comparison - 1BM24MC054

- **Problem Statement:** A client is confused about IaaS, PaaS, and SaaS. You must analyze and recommend suitable models.
- **Tasks:**
  1. Research examples of IaaS, PaaS, SaaS.
  2. Write a case study (e.g., e-commerce using Shopify vs AWS EC2).
- **Expected Output:** 2–3 page report.

Lab 4: Create and Configure a VM - 1BM24MC036

- **Problem Statement:** A company needs a cloud server for its internal tools.
- **Tasks:**
  1. Launch EC2/Compute Engine VM.
  2. Connect via SSH.
  3. Install basic software (e.g., text editor, curl).

Lab 5: Install Web Server - 1BM24MC045

- **Problem Statement:** A small business needs a website hosted in the cloud.
- **Tasks:**
  1. Deploy Apache/Nginx on VM.
  2. Place a sample index.html.
  3. Access via public IP.

Lab 6: Deploy a Database VM - 1BM24MC041

- **Problem Statement:** A company needs cloud-based customer database access.
- **Tasks:**
  1. Launch a VM with MySQL/PostgreSQL.
  2. Create a sample database + table.

3. Insert & retrieve records.

Lab 7: Cloud Storage Demo - 1BM24MC048

- **Problem Statement:** A college needs a secure place to store and share documents.
- **Tasks:**
    1. Create a cloud storage bucket.
    2. Upload different file types.
    3. Share via pre-signed URL/public link.

Lab 8: Versioning in Cloud Storage - 1BM24MC025

- **Problem Statement:** A design firm wants to track changes in their project files.
- **Tasks:**
    1. Enable bucket versioning.
    2. Upload & overwrite a file multiple time.
    3. Restore old version.

Lab 9: SaaS Migration Case Study - 1BM24MC023

- **Problem Statement:** A company uses Excel-based HR system and wants SaaS migration.
- **Tasks:**
    1. Identify SaaS alternatives (Workday, Zoho People, etc.).
    2. Write benefits vs risks.
    3. Recommend best solution.

Lab 10: Cost Estimation Exercise - 1BM24MC014

- **Problem Statement:** A client wants to know cost of running a 3-tier app on cloud.
- **Tasks:**
    1. Use AWS/Azure calculator.
    2. Estimate cost of web + app + DB servers.
    3. Suggest optimization (reserved/spot).

Lab 11: Create Multiple VMs - 1BM24MC027

- **Problem Statement:** A company needs three servers (web, app, database) in the cloud.
- **Tasks:**
    1. Launch 3 VMs in the same VPC/subnet.
    2. Assign roles (Web/App/DB).
    3. Verify internal communication using ping/SSH.

Lab 12: Load Balancer Setup

- **Problem Statement:** A shopping site needs high availability for its web server.
- **Tasks:**
    1. Launch 2 web servers with sample webpages.
    2. Configure a cloud load balancer.
    3. Test traffic distribution by refreshing browser.

Lab 13: Auto-Scaling Demo - 1BM24MC004

- **Problem Statement:** During a flash sale, traffic spikes on an e-commerce site.
- **Tasks:**
    1. Configure an Auto-Scaling Group with 2–4 instances.
    2. Define scaling policy (CPU > 60%).
    3. Test by simulating load.

Lab 14: VPC Configuration - 1BM24MC013

- **Problem Statement:** A company wants isolated networking for its app.
- **Tasks:**
    1. Create a VPC with public and private subnets.
    2. Launch VMs in each subnet.
    3. Configure routing rules.

Lab 15: Firewall & Security Groups

- **Problem Statement:** A bank wants only port 443 open for a secure website.
- **Tasks:**
    1. Launch VM with a web server.
    2. Create security group allowing only HTTPS (443).
    3. Attempt connections on blocked ports.

Lab 16: VPN Connection

- **Problem Statement:** A company wants secure access from office to cloud servers.
- **Tasks:**
    1. Set up a VPN gateway in cloud.
    2. Configure VPN client locally.
    3. Test secure access to private VM.

Lab 17: Hybrid Cloud Demo - 1BM24MC049

- **Problem Statement:** A company wants to keep database on-premise but app on cloud.
- **Tasks:**
    1. Run a DB locally (MySQL/Postgres).
    2. Launch a cloud VM running a sample app.
    3. Connect cloud app to local DB.

Lab 18: Banking App Network Security - 1BM24MC017

- **Problem Statement:** A bank needs a secure cloud design for online banking.
- **Tasks:**
    1. Design 3-tier architecture (Web/App/DB).
    2. Place DB in private subnet, App in middle, Web in public subnet.
    3. Define firewall rules.

Lab 19: Elastic IP Assignment

- **Problem Statement:** A company wants a permanent IP for their cloud server.
- **Tasks:**
    1. Launch a cloud VM.
    2. Allocate an Elastic IP.
    3. Associate IP with VM.

Lab 20: DNS Mapping

- **Problem Statement:** A startup wants to map their domain (myshop.com) to a cloud website.
- **Tasks:**
    1. Register/purchase a domain (use free DNS if possible).
    2. Create DNS record pointing to cloud server IP.
    3. Verify by opening domain in browser.

Lab 21: Serverless Function - 1BM24MC057

- **Problem Statement:** A news portal wants an automatic image resizer when reporters upload photos.
- **Tasks:**
    1. Create an AWS Lambda/Azure Function/Cloud Function.
    2. Trigger on file upload to storage bucket.
    3. Resize image and save processed copy.

Lab 22: API Deployment

- **Problem Statement:** A company wants to expose employee data via a secure API.
- **Tasks:**
    1. Use API Gateway service.
    2. Connect API to a Lambda/Function backend.
    3. Test with GET/POST requests.

Lab 23: Chatbot on Cloud

- **Problem Statement:** A retail store wants a chatbot to answer customer queries.
- **Tasks:**
    1. Use AWS Lex/Google Dialogflow/Azure Bot Service.
    2. Create intents (greetings, store hours, product query).
    3. Deploy bot on a web page or test console.

Lab 24: Online Exam Portal

- **Problem Statement:** A university wants a cloud-based system for online exams.
- **Tasks:**
    1. Deploy web app on VM/Cloud App Service.
    2. Store questions/answers in cloud database.
    3. Enable login for students.

Lab 25: E-Commerce Case Study

- **Problem Statement:** A startup wants to migrate their online shop to cloud.
- **Tasks:**
    1. Design 3-tier cloud architecture (web/app/db).
    2. Propose scaling and security measures.
    3. Provide cost estimate.

Lab 26: IoT Integration – 1BM24MC044

- **Problem Statement:** A smart home company wants to collect sensor data in the cloud.
- **Tasks:**
    1. Simulate IoT device using MQTT client.
    2. Connect device to AWS IoT Core/Azure IoT Hub.
    3. Visualize data in dashboard/DB.

Lab 27: Stream Processing

- **Problem Statement:** A news app needs to process live tweets about elections.
- **Tasks:**
    1. Collect streaming data (use sample dataset or simulated logs).
    2. Process using AWS Kinesis/Google Dataflow.
    3. Store results in a cloud DB.

Lab 28: Big Data Analysis

- **Problem Statement:** A healthcare agency wants to analyze patient health dataset.
- **Tasks:**
    1. Launch Hadoop/Spark cluster in cloud (EMR/Dataproc/HDInsight).
    2. Run queries (count, filter, group).
    3. Generate basic analytics report.

Lab 29: AI/ML Deployment

- **Problem Statement:** A bank wants to predict loan approval using ML in the cloud.
- **Tasks:**
    1. Use AWS SageMaker/Azure ML/Vertex AI.
    2. Train a simple classification model on sample dataset.
    3. Deploy model as endpoint.

Lab 30: Smart City IoT Case Study

- **Problem Statement:** A smart city needs real-time traffic and pollution monitoring.
- **Tasks:**
    1. Propose architecture using IoT + Cloud Analytics.
    2. Include data ingestion, processing, storage, visualization.
    3. Suggest scaling & security solutions.

Lab 31: Identity & Access Management

- **Problem Statement:** A company wants controlled access for its employees in the cloud.
- **Tasks:**
    1. Create cloud IAM users (Admin, Developer, Auditor).
    2. Assign least-privilege roles to each.
    3. Test access restrictions.

Lab 32: Multi-Factor Authentication

- **Problem Statement:** A financial firm wants to strengthen login security.
- **Tasks:**
    1. Enable MFA for cloud account.
    2. Configure authenticator app/OTP device.
    3. Demonstrate MFA-enabled login.

Lab 33: Encrypt Data at Rest

- **Problem Statement:** A hospital must encrypt patient records in cloud storage.
- **Tasks:**
    1. Create encrypted storage bucket or encrypted database.
    2. Upload confidential data.
    3. Show encryption settings enabled.

Lab 34: Encrypt Data in Transit

- **Problem Statement:** An e-commerce site wants HTTPS-enabled transactions.
- **Tasks:**
    1. Configure SSL/TLS certificate for a cloud web server.
    2. Redirect all HTTP traffic to HTTPS.
    3. Verify secure connection in browser.

Lab 35: Disaster Recovery Setup

- **Problem Statement:** A university wants backups for its student database in case of failure.
- **Tasks:**
    1. Backup database to a different region.
    2. Simulate primary failure.
    3. Restore backup to new VM/DB instance.

Lab 36: Monitoring & Logging

- **Problem Statement:** A company wants to track server health and activity.
- **Tasks:**
    1. Enable CloudWatch/Stackdriver monitoring for a VM.
    2. Track CPU, memory, and network usage.
    3. Set an alert for CPU > 70%.

Lab 37: Cost Optimization

- **Problem Statement:** A startup is paying too much for cloud resources.
- **Tasks:**
    1. Analyze usage (running VMs, storage, networking).
    2. Identify unused/underutilized resources.
    3. Suggest cost-saving options (reserved, spot, auto-scaling).

Lab 38: Compliance Case Study

- **Problem Statement:** A healthcare provider must follow HIPAA for cloud data storage.
- **Tasks:**
    1. Research compliance requirements (HIPAA/PCI DSS/GDPR).
    2. Map how cloud provider services meet compliance.
    3. Write a compliance checklist.

Lab 39: Multi-Cloud Deployment

- **Problem Statement:** A company wants redundancy using AWS + Azure.
- **Tasks:**
    1. Deploy a sample app on AWS + Azure.
    2. Configure DNS to switch traffic between them.
    3. Compare cost/performance.

Lab 40: Capstone Project (Choose One)

Problem Statement: A client needs a complete cloud solution. Choose a scenario:

- E-Commerce Store
- Online Exam System
- Video Streaming Platform
- Healthcare Patient Portal

Tasks:

1. Design end-to-end cloud architecture (networking, storage, compute, security).
2. Deploy a working prototype (even if partial).
3. Prepare final presentation/report.

41. Launch a Virtual Machine (EC2 / VM Instance)

- Spin up a small Linux/Windows instance.
- Connect via SSH (Linux) or RDP (Windows).
- Try starting, stopping, and terminating the instance
- *Goal:* Learn how to provision and manage basic compute resources.

42. Create and Attach Storage

- Create a **block storage volume** (EBS in AWS, Disk in Azure).
- Attach it to a VM.
- Format, mount, and use it.
- *Goal:* Understand persistent storage basics.

43. Set Up a Simple Static Website

- Create an **S3 bucket / Blob Storage**.
- Enable **static website hosting**.
- Upload an index.html file.
- Access it using the bucket's endpoint URL.
- *Goal:* Learn about object storage and static hosting.

44. Work with Security Groups / Firewall Rules

- Launch a VM and restrict inbound access.
- Allow only your IP for SSH/RDP.
- Try accessing from an unauthorized IP to test.
- *Goal:* Practice basic cloud security.

45. Create a Load Balancer

- Launch two small VMs running a simple web server (Apache/Nginx).
- Put them behind a **Load Balancer (ELB in AWS, ALB in Azure)**.
- Access the LB DNS name and see traffic distribute between instances.
- *Goal:* Learn high availability and traffic distribution.

46. Try Auto Scaling

- Create an **Auto Scaling Group** with a launch template.
- Define scaling policies (e.g., add instance when CPU > 70%).
- Test by increasing load.
- *Goal:* Understand elasticity in the cloud.

47. Create and Use a Database Service

- Launch a managed database (e.g., AWS RDS, Azure SQL, Cloud SQL).
- Connect using a client tool or VM.
- Perform basic queries.
- *Goal:* Learn managed database concepts.

48. Set Up Cloud Monitoring

- Enable **CloudWatch (AWS) / Monitor (Azure) / Stackdriver (GCP)**.
- Monitor CPU usage of a VM.
- Set up an alarm to send an email when CPU > 70%.
- *Goal:* Learn cloud monitoring and alerting.

49. Experiment with IAM (Identity and Access Management)

- Create a new user with limited permissions.
- Log in with that user and test restrictions.
- *Goal:* Understand least-privilege access.

50. Create a Backup & Restore

- Take a **snapshot** of a VM volume.
- Terminate the VM.
- Launch a new VM using the snapshot.
- *Goal:* Learn disaster recovery basics.

51. Create a Read-Only IAM User – assign S3 Read Only Access and test permissions.

52. Deploy a Free SSL Certificate – use AWS Certificate Manager and attach to a Load Balancer.

53. Launch a Linux and Windows VM – compare SSH vs RDP login.